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THE EFFECT OF CHILDCARE ON FEMALE LABOR SUPPLY
AND CHILD DEVELOPMENT IN LOW AND MIDDLE INCOME COUNTRIES

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The Effect of Childcare on Female Labor Supply and Child Development in Low and Middle Income Countries

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ABSTRACT

Childcare services can offer an opportunity for women to develop their careers and increase household income. At the same time, it can affect the development of children, depending on the quality of care. We first develop a collective model of the household as a conceptual framework to show how improving the availability of childcare can affect mothers' labor market opportunities and earnings and how it can affect child development. We then review the literature on the effects of childcare on the employment and earnings of women and on child development. Childcare can increase mothers' employment. However, limited work opportunities or strict norms against maternal work can keep mothers out of the labor force. When other family members provide care, childcare can free these caregivers to work or attend school. When studies measure the impact on child development, the results are mixed and depend on both context and quality of provision.

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An online appendix is available at <http://www.nber.org/data-appendix/w35649>

1 Introduction

Women are 24pp less likely to participate in the labor force than men worldwide ([World Bank, 2025](#)). Labor market conditions, social norms, and childcare availability are all candidate explanations. Given the number of women who do not work in the labor market, increasing women’s labor force participation can have transformational impacts on households’ standard of living. Beyond this simple observation, work, or even the option of taking a job, can empower women by improving outside options and access to income ([Molina and Tanaka, 2023](#)), sometimes fundamentally altering her control of household resources. Women who control more resources can invest more in their children, leading to better child outcomes ([Blundell, Chiappori, and Meghir, 2005](#)) and fostering a new generation of empowered women; expanded opportunities for women can also transform girls’ prospects directly. In Bangladesh, growth of the garment industry raised girls’ schooling and delayed marriage through the anticipated returns to education, even in families where no woman worked in the sector ([Heath and Mobarak, 2015](#)).

In this paper, we review the effects of providing childcare as an approach to improve women’s participation in the labor market. By childcare, we mean center-based formal care for children of pre-school age. The key question is whether such a policy can be used for the dual purpose of improving women’s empowerment and addressing the deficits in child development pervasive in developing countries and among poor populations more generally. Childcare infrastructure in Low and Middle Income Countries (LMICs) is often underdeveloped, leaving mothers with limited time and children without stimulation ([Devercelli and Beaton-Day, 2020](#)). The literature points to key conditions for success, such as quality of

care and existing opportunities in the labor market for women, but also remaining gaps.¹

First, we develop as a conceptual framework a collective household model ([Chiappori, 1988, 1992](#)) under full information that includes the production of child human capital, to determine the conditions under which lowering the price of childcare increases maternal employment and improves children’s development. The model is useful because it illustrates the potential complexity of responses to such a policy and the multiple alternative impacts it may have, even before we consider complications such as social norms and beliefs about child development.

According to this model, when a mother already uses more childcare than hours she works, lowering the cost of childcare has an ambiguous effect on hours. By lowering the marginal cost to produce child human capital, childcare subsidies increase the household’s demand for child human capital. Households then reoptimize inputs to achieve the cheapest way of increasing child human capital to the new optimum, which may involve increasing maternal time input if this is complementary to childcare. This is the standard separation of production from preferences ([Becker, 1965](#); [Gronau, 1977](#); [Singh, Squire, and Strauss, 1986](#); [Benjamin, 1992](#)). In this case, a childcare subsidy will improve child development, but may increase or decrease the mother’s hours of work depending on the structure of the human capital production function. When the reduction in costs also strengthens the mother’s bargaining position, the effect on child human capital is reinforced if her marginal willingness to pay for child human capital responds more strongly to her private resources than the father’s does to his own ([Blundell, Chiappori, and Meghir, 2005](#), BCM). The effect

¹See also [Evans, Jakiela, and Knauer \(2021\)](#), which quantify the number of articles in this area and describe the sizes of the effects of childcare on mothers, especially on their labor supply, mental health, and decision-making power, in the literature.

on hours remains ambiguous however.

The implications are more complex if the household uses childcare for only the hours the mother works (the constrained optimum where the household must use childcare for the hours the mother works, but would use less childcare if they could). A childcare subsidy may harm child development if it induces the mother to work longer hours and is below a quality threshold. Importantly for the policy discussion, this also applies when mothers could not work at baseline because of the lack of (affordable) childcare and the subsidy pushes her to enter.

Households may start using childcare after receiving a subsidy even when it will reduce child human capital if there are sufficiently attractive labor market opportunities. Higher-quality childcare can further reduce the costs of using childcare, encouraging take-up and reducing the costs of maternal work, especially if high-quality care is a substitute for maternal time. This point illustrates the central importance of childcare quality if we are to achieve the dual aim of supporting women's careers and promoting child development. In addition to the standard effects operating through prices, there is also a possible bargaining power channel that can further shift resources to the child under conditions we discuss.

Thus, there are multiple channels that can determine the impact of childcare availability on women's work and child development. This includes first and foremost the quality of childcare. But the relative preferences of father and mother, the determination of their relative bargaining power, and the structure of the human capital production function are key factors. Ultimately, the impact must be determined empirically. Understanding the mechanisms requires empirically estimated models of the household and of the human capital production function for children, based on credible research designs.

Next, we consider the evidence from the literature, which introduces many more nuances and dimensions than those we capture in the model, such as informal care from grandparents and siblings, the importance of social norms and the availability of labor market opportunities for women. If a mother already use informal care, subsidized care has a smaller effect on her labor supply. However, these caregivers, such as grandparents or older siblings, may then begin to work or stay in school. On the other hand, if mothers are generally their children’s primary caregivers, an offer of childcare can increase maternal employment and hours. To encourage mothers to join the labor force, childcare must cover a large enough part of the day and must be coordinated with existing employment opportunities for women. If norms, discrimination, or poor economic conditions limit potential earnings, freeing mothers’ time may not be enough. Higher-quality care may reduce maternal labor supply while increasing parental investment, perhaps by making maternal time more productive for child human capital.

If children already use high-quality care, subsidizing lower quality public childcare can harm their development, particularly if care time is tied to hours of work. When the alternative is simultaneous care while working (say, in the home business), quality childcare may make mothers’ labor market time more productive and give children much needed stimulation. Preschool has mixed effects on children’s longer-term academic results, but this can be related to both low quality ([Baker, Gruber, and Milligan, 2008](#)) and other factors such as complementary investments in later education ([Johnson and Jackson, 2019](#)).

This paper proceeds as follows. Section 2 presents the model and discusses implications. Section 3 reviews the literature and evaluates how well it lines up with the model’s predictions. Section 4 discusses obstacles to the success of childcare interventions. Section 5

discusses and concludes.

2 Model and Mechanisms

In this section, we use a model to organize a discussion of the basic channels through which a childcare intervention (such as reducing the cost, or increasing quality) may affect household decisions on the allocation of time and resources devoted to child human capital. We build on the household production model of [Becker \(1965\)](#); [Gronau \(1977\)](#); [Benjamin \(1992\)](#) combined with the collective model of [Chiappori \(1988, 1992\)](#) as extended by [Blundell, Chiappori, and Meghir \(2005\)](#), henceforth BCM) to include public goods (child human capital here). The collective model allows an explicit treatment of the role of women’s empowerment, which here will be represented by their Pareto weight, which drives her influence on household decisions.

2.1 A collective model of the household

Human Capital Production Parents invest in their children by providing active time (τ^m and τ^f), and expenditure e . They can also purchase formal childcare for d hours, with quality adjusted hours $\kappa(q)d$, where $\kappa(q) > 0$ and $\kappa'(q) > 0$. The human capital/skill production function is

$$\theta = g(\tau^m, \tau^f, e, \kappa(q)d), \quad (1)$$

We assume g is increasing and concave in all arguments, which implies childcare is not so low quality that increasing it without reducing other inputs harms the child.² Implicit in what

²We simplify matters by ignoring passive time at home, which in practice may not add much, but may

follows is the assumption that parents know the technology of skill formation $g(\cdot)$. This is a strong assumption, which we discuss below (Cunha, Elo, and Culhane, 2022).

We assume that childcare hours must at least cover hours of work, but they can also be higher $d \geq h^f$.³

Collective Household Problem Each parent $j \in \{m, f\}$, standing for mother and father respectively, has preferences represented by an additively separable utility function

$$U^j(c^j, \ell^j, \theta) = \phi^j(c^j) + \psi^j(\ell^j) + \xi^j(\theta), \quad j \in \{m, f\}, \quad (2)$$

where c^j denotes private consumption of parent j , ℓ^j denotes the leisure of parent j , and θ denotes the human capital of the child, which enters parents' utility and is a public good. Each component is increasing and strictly concave.⁴

Under the collective approach, the household maximizes a weighted sum of utilities (Pareto efficiency). The household solves:

$$\max_{\substack{c^m, c^f, h^m, h^f, \\ \tau^m, \tau^f, e, d}} \mu U^f(c^f, \ell^f, \theta) + (1 - \mu) U^m(c^m, \ell^m, \theta) \quad (3)$$

be better than lowest-quality childcare centers.

³In the model, we do not consider unpaid care by other family members or neighbors.

⁴Additive separability is not an innocuous assumption, since it imposes substitutability between all goods. Many of the results we present do not depend on it, but more complex analyses with corner solutions become simpler to derive.

subject to:

$$\theta = g(\tau^m, \tau^f, e, \kappa(q) d), \quad (4)$$

$$c^m + c^f + e + (1 - s)p(q) d = w^m h^m + w^f h^f + y, \quad (5)$$

$$h^j + \tau^j + \ell^j = T, \quad j \in \{m, f\}, \quad (6)$$

$$d \geq h^f, \quad (7)$$

$$c^j \geq 0, \quad h^j \geq 0, \quad \tau^j \geq 0, \quad \ell^j \geq 0, \quad j \in \{m, f\}, \quad (8)$$

$$e \geq 0, \quad d \geq 0. \quad (9)$$

where, w^j is the market wage of parent j , y is non-labor income and $\tilde{p} \equiv (1 - s)p(q)$ is the price of childcare of (given) quality q subsidized at rate s . Bargaining power, as reflected by the Pareto weight $\mu = \mu(w^m, w^f, y, s) \in (0, 1)$ may depend on the subsidy s , because it affects the outside option (as well as utility within the marriage).

The first order conditions for the mother's time investment in child human capital and for childcare hours imply:⁵

$$\frac{\Omega g_2}{\mu U_c^f} = w_f - \varphi^* \quad (10)$$

$$\frac{\Omega [\kappa(q) g_4]}{\mu U_c^f} = (1 - s)p(q) - \varphi^* \quad (11)$$

⁵The solution of the model and some comparative static results are presented in the online appendix.

where $\Omega \equiv \mu U_\theta^f + (1 - \mu) U_\theta^m$ is the marginal value of the public good to the household, and where $\varphi^* = \varphi/(\mu U_c^f)$ is the shadow cost of the minimum childcare constraint, which, in general will depend on both preferences and the Pareto weight. The first condition equates the marginal product of the mother's time input for child human capital (g_2 , scaled by the joint willingness to pay, $\Omega/\mu U_c^f = \text{MWP}^f + \text{MWP}^m$) to the mother's wage net of φ^* , and similarly for childcare hours for the second condition. Combining the two, we get

$$\frac{g_2}{\kappa(q) g_4} = \frac{w_f - \varphi^*}{(1 - s)p(q) - \varphi^*} \quad (12)$$

which implies that in the absence of corner solutions ($\varphi^* = 0$), the optimal allocation between mother's time and childcare only depends on the technology of skill formation (g_2 and g_4) and prices (w_f and $(1 - s)p(q)$). Given the household's decision on leisure and the desired level of θ , neither preferences nor bargaining power enter the determination of how non-leisure time is allocated between the market and child human capital. This is just an optimal production decision (Becker, 1965; Gronau, 1977). Importantly, if childcare time is constrained by hours of work ($d = h^f$), this result breaks down, and allocations to the production of child human capital depend on bargaining power μ and preferences through φ^* .

Child human capital. As in BCM, parents choose the optimal level of human capital based on the BLS condition (Bowen, 1943; Lindahl, 1958; Samuelson, 1954)

$$\text{MWP}^f(\rho^f, \theta) + \text{MWP}^m(\rho^m, \theta) = C_\theta, \quad (13)$$

where C_θ is the marginal cost of producing θ and the marginal willingness to pay is defined by

$$\text{MWP}^i(\rho^i, \theta) \equiv \frac{\partial v^i / \partial \theta}{\partial v^i / \partial \rho^i}, \quad i \in \{f, m\}. \quad (14)$$

Here, $v^i = v^i(\rho^i, \theta)$ is the conditional indirect utility function where private decisions are a function of individual resources (ρ^i) and wages. The marginal willingness to pay of parent i (MWP^i) is decreasing in child human capital and increasing in their share of resources (θ is a normal good). We suppressed prices from the notation for simplicity.⁶

Using this setup and in the absence of corner solutions [Blundell, Chiappori, and Meghir \(2005\)](#) derive the BCM condition

$$\frac{\partial \theta}{\partial \mu} > 0 \quad \Longleftrightarrow \quad \Delta \text{MWP}_\rho \equiv \frac{\partial \text{MWP}^f}{\partial \rho^f} - \frac{\partial \text{MWP}^m}{\partial \rho^m} > 0. \quad (15)$$

In other words, an increase in mother's bargaining power increases the desired child human capital when the mother's marginal willingness to pay for θ with respect to her share of resources, rises faster than the father's with respect to his share. The intuition is simple: the shift in resources to the mother must increase her marginal willingness to pay, more than it decreases his if the demand for the public good (child human capital here) is to increase. In this case, the demand for child human capital rises and with it all (normal) inputs such as time with the child, expenditures, and possibly childcare if the price is not too high and we

⁶In the analysis of corner solutions, when hours of work drive the amount of childcare used, the mother's leisure is priced at her shadow wage $\tilde{w}^f$, which is endogenous and moves with subsidies and bargaining power. The mathematical analysis involving corner solutions is greatly simplified by assuming additive separability of preferences. The shadow wage then affects the allocation of the mother's time through household resources and the availability, price and quality of childcare, not through the form of MWP^f .

are not at a corner solution of $d = 0$.

There are some interesting special cases. For quasi-linear preferences, bargaining power has no effect on human capital because there are no income effects. And for Cobb Douglas preferences, the condition $\Delta \text{MWP}_\rho > 0$ holds when the marginal willingness to pay of the mother is higher than the father's.⁷ In what follows, we will assume that $\Delta \text{MWP}_\rho > 0$.⁸

2.2 The impact of childcare subsidies

We can think of programs that expand childcare availability as a cost reduction caused, say, by reducing the distance to a center.⁹ The subsidy can affect household choices directly because it affects a price of one of the inputs. However, it can also change household choices by shifting the bargaining power, reflected in the Pareto weight of each parent. In what follows, we describe the key implications of the model.¹⁰

We consider two cases. In the first, households are at an interior optimum before the childcare subsidy is increased: both parents work and use childcare in excess of mother's hours worked ($h^m > 0$, $h^f > 0$, $d > 0$, $d > h^f$). In the second, we address the important case of households where the mother initially neither works nor uses childcare.

For interior solutions, subsidizing childcare raises the demand for child human capital unambiguously for fixed bargaining power. This is because by subsidizing the price of childcare, an input child human capital becomes cheaper to produce. Hence as the marginal cost of producing child human capital falls, the household demands more of it and increases all

⁷With Cobb-Douglas utility each MWP^i is linear in ρ^i , so the comparison of slopes reduces to a comparison of levels at equal ρ^i .

⁸We provide greater mathematical detail for the model and the comparative statics in an online Appendix.

⁹We consider an unconditional subsidy. One can think of policies designed to incentivize both work and childcare use by conditioning the subsidy on the mother working.

¹⁰The underlying mathematical treatment can be found in the Appendix.

inputs in the child human capital production function based on relative prices and productivities. If the Pareto weight also shifts in favor of the mother, this result is reinforced, so long as the BCM condition holds and $\partial\mu/\partial s > 0$. If it does not, then the outcome is ambiguous. Note that this result is valid irrespective of childcare quality, under the assumption of full information, something we discuss below.

Effect on female labor supply. The effect of the subsidy on hours of work is more complex. The subsidy is not a direct work incentive. There are three channels that combine to define the effect. First, there is a positive effect on hours of work if her time with the child τ^f and childcare time d are substitutes in the human capital production function. Higher-quality childcare can strengthen the indirect work incentives of the subsidy. Working in the opposite direction is the increase in the demand for child human capital which demands more of the mother's time input τ^f . Finally, the subsidy increases household resources (because of the reduction in cost of childcare the household would have used regardless) and moreover, potentially increases the mother's Pareto weight, shifting resources to the mother. With greater resources, she may increase her demand for leisure, as well as possibly the demand for child human capital under BCM. Thus, at least in theory, a subsidy could reduce mothers' labor supply while still increasing child human capital.

Subsidies and take-up of childcare and work Consider a household where the mother does not work and no childcare is used ($h^f = 0$ and $d = 0$). The household can find itself in that position for two different reasons and the distinction matters.¹¹ In the first, her labor

¹¹We always assume that leisure is not at a corner and that there is always a positive amount of time to be allocated between taking care of the child and working.

market opportunities and wage are such that she does not want to work; the availability of childcare is not a barrier. In the second, she would want to work at her wage ($g_2/g_3 < w^f$), but working requires childcare that is too expensive and the availability of childcare becomes the barrier to work.

In the first case, as the subsidy rises, only the childcare condition is affected and her incentive to work is unchanged so long as the household continues to not buy childcare. If the subsidy is large enough that the household starts to take up childcare even as the mother remains at home then the analysis of the previous section applies: child human capital rises unambiguously with the subsidy once parents start using it. The mother will start working if her marginal productivity in child human capital production declines sufficiently, which may happen if childcare is a good substitute for the mother's time input. Thus, the higher the quality of the available childcare, the greater the incentive for the mother to work. Hence, in the context of this full information world, when a subsidy induces the use of childcare it improves child human capital irrespective of quality, because it raises the demand for child human capital, which is achieved by a reallocation of inputs. However, higher-quality care can improve the mother's work incentives relative to taking care of the child.

In the second case, she wants to work, but the lack of affordable childcare prevents her. At a high enough subsidy, the mother enters the labor market, with childcare covering exactly her hours of work with the constraint $d = h^f$ binding. The minimum necessary subsidy is smaller if her labor market opportunities are stronger (e.g., higher w^f) and depends on the shape of the child human capital function. If higher-quality care reduces the effectiveness of maternal time and expenditures, acting as a substitute, higher-quality care unambiguously induces entry at a smaller subsidy if the gains from formal childcare, net of the losses of

maternal time, are positive at entry. The latter holds when childcare is higher-quality relative to counterfactual care. If maternal time inputs are more effective at higher-quality care, however, the effect of quality of care on hours of work is ambiguous.

Shutting down for now the bargaining power channel ($\partial\mu/\partial s = 0$), a simple expression summarizing succinctly the effects of an increase in hours of work and a resulting equal increase of formal childcare is given by

$$\left. \frac{d\theta}{dh^f} \right|_{d=h^f} = \kappa(q) g_4 (1 - \beta) + g_2 (\beta - \alpha). \quad (16)$$

where $\kappa(q)g_4$ is the marginal product of an hour of childcare, while g_2 is the productivity of the mother's hours. In this expression, β is the share of net marginal labor income ($w^f - (1 - s)p(q)$) allocated to θ , while α is the share of mother's time taken away from the child as a result of her increased labor supply (the rest comes from a reduction in leisure).¹²

The first term in (16) is strictly positive when childcare contributes to child development ($\kappa(q)g_4 > 0$). The second term is also positive if $\beta \geq \alpha$, i.e. if the share of marginal resources directed to the child is at least as large as the share of the hours drawn from her time with the child because of her labor market entry, in which case the direct effect is unconditionally positive. However, if $\beta < \alpha$, the direct effect becomes positive at a threshold quality q^* defined by

$$\kappa(q^*) g_4 = \frac{\alpha - \beta}{1 - \beta} g_2, \quad (17)$$

where the impact on child human capital is positive only above the quality threshold q^* . The

¹²We assume that $\alpha, \beta \in (0, 1)$. These are endogenous functions of prices and resources, in general.

implication is that when childcare is a barrier to mothers working and the amount used is tied to her hours of work, child development can be harmed by a subsidy unless the quality of the available care is sufficiently high.

Effect of the bargaining channel. Childcare availability can shift intrahousehold bargaining power in favor of mothers, changing intrahousehold allocations in her favor. In the unconstrained case ($d > h^f$) the BCM condition $\Delta MWP_\rho > 0$ implies an increase in demand for child human capital. However, when childcare is tied to hours of work, this clean result no longer holds. In this case, the bargaining effect adds ambiguity to the impact on child human capital, but care still needs to be of minimum quality to improve children’s human capital.

Discussion Offering households high-quality care may not increase maternal labor supply, but instead may engage parents more actively in their child’s development, by raising their demand for child human capital and the value of the mother’s time with children. Consistent with this, [Hojman and Lopez Boo \(2022\)](#) find that higher-quality childcare raises parental investments more than low-quality childcare centers.

When childcare acts as a barrier to mothers working, subsidizing childcare may alleviate this barrier and allow them to enter the labor market. If that happens, the effect on child human capital depends on the quality of care provided, with low-quality care harming child development. On the other hand, the cheap provision of quality childcare is a win-win policy, increasing mothers’ work potential and improving child development.

The declines in human capital as a result of the increased availability of childcare are

unfortunately realistic and have been well documented in Canada ([Baker, Gruber, and Miligan, 2008](#)). On the other hand, there are counterexamples: [Almås, Drange, Meghir, and Zachrisson \(2026\)](#) show that early attendance at childcare can have substantial positive impacts for children from disadvantaged backgrounds in the context of high-quality provision, such as in Norway. Sufficient levels of quality must be ensured to avoid the benefit from mothers' employment translating to child development deficits.

Outcomes are relative to a counterfactual, which in our model is reflected in the productivity of home inputs and expenditures for children and in her labor market opportunities and wage. She may, for example, be restricted from working outside the home, but working on the family farm while taking care of her children, which may not be very stimulating for the child. Increased availability of childcare can then increase her productivity on the farm ([Delecourt and Fitzpatrick, 2021](#); [Donald, Lowes, and Vaillant, 2024](#)), while giving the child relatively stimulating childcare. Although this may not register as an increase in hours of work, the return to such informal unpaid jobs increases and child development may improve ([Donald, Lowes, and Vaillant, 2024](#)). On the other hand, this does not necessarily improve women's control of resources ([Halim, Johnson, and Perova, 2022](#)), and may have little or no effect on women's empowerment.

More generally, whether work itself improves women's empowerment is questionable. Working in the market or the household business may not improve her position if the husband can control the use of her earnings. However, institutional changes that increase her options and ability to control household expenditures can be a genuine source of empowerment, and the broader availability of childcare may in part achieve this. More broadly, a shift in the conditions and norms that allow women greater freedom of movement is likely to broaden

their social contacts ([Andrew, Attanasio, Augsburg *et al.*, 2020](#)), improve their welfare, and empower them, even if they do not take up the option to work. And increased empowerment can also positively affect the children, as expressed formally in the model by the BCM condition.

The presence and use of grandparents or older siblings, and in particular girls, for watching over a younger child can change whether maternal employment responds to improving the availability of formal childcare. Women may have been able to work before, passing on the responsibilities of housework and childcare to them. Once formal childcare becomes available, grandparents may be released from these duties, and older siblings may return to school or work, without seeing much change in mothers' labor supply. However, the availability of childcare gives more control to the mother and improves her well-being, which can impact the way the household allocates resources. These are more subtle effects, but no less important because, for example, they can improve schooling outcomes for older children.

Bargaining Power and Childcare Availability. In our model, one of the channels through which childcare can affect women's labor supply and child development is the influence of the mother on household decisions, which we formally define as bargaining power.

To better understand how this channel operates, we need to ask how Pareto weights are determined in the first place. [Chiappori, Costa-Dias, and Meghir \(2018\)](#) present a model in which the Pareto weight that governs the distribution of intrahousehold resources is determined at the time of a match (*ex ante*) to ensure the marriage market clears. The Pareto weight depends on the value of alternative matches and of being single. Social norms play a key role here. For example, childcare infrastructure is likely to play a more important

role in determining women’s relative bargaining power at the time of marriage if being a single mother is socially acceptable.

Once married, under a full commitment regime, Pareto weights do not change with improved outside options induced, say, by new childcare infrastructure. However, if commitment is limited, improving outside options produces a credible threat to divorce and can shift bargaining power and resources from one spouse to another ([Mazzocco, 2007](#)). In a seminal contribution demonstrating the importance of the institutional context, [Voena \(2015\)](#) argues that mutual-consent divorce regimes can sustain full commitment within marriage, while unilateral divorce makes exit credible and thereby enables renegotiation. In this case, changes in the value of the outside option, resulting from improvements in childcare infrastructure, can trigger renegotiation and a shift of resources to the mother.

In certain cultural contexts, divorce may not be a credible threat (even if legally feasible). However, this does not always mean that renegotiation is not possible because of other threat points such as moving to a noncooperative equilibrium within an intact marriage, as in the separate spheres model of [Lundberg and Pollak \(1993\)](#). This channel can have its limits though, when norms also restrict married mothers from working. This does not mean that investments in child human capital will not be affected by a broader availability of childcare even in such settings. The key channels we have identified for these effects are the cost of producing child human capital and the choice of parental inputs. Moreover, we have already shown that a shift of resources to the mother does not guarantee more investments in children.

Finally, we have assumed that the availability of childcare improves women’s bargaining power. But this may not be the case necessarily, especially in contexts where women’s choices

are restricted. For example, with formal childcare mothers are no longer the sole providers of care, which might weaken their position. Nevertheless, the little evidence we have points to the improvement of women’s bargaining power with broader childcare availability ([Chiappori, Meghir, and Okuyama, 2025](#)).

Parental beliefs The discussion has assumed that parents know the skill production function and the trade-offs between inputs. Predicting how behavior is determined when this is not true is particularly complex. [Cunha, Elo, and Culhane \(2022\)](#) show that low-income parents misperceive the production function of human capital, leading to under-investments in children. Although the above study refers to low-income women in Philadelphia, similar conclusions are drawn in the context of Colombia by [Attanasio, Cunha, and Jervis \(2019\)](#).

More broadly, parents may have distorted beliefs about the role of nutrition, early parenting, and of childcare quality and its importance, generating a chain of under-investments from the very early days and even before birth. Making childcare more available in that context without adequate levels of quality and appropriate interventions to improve child outcomes can lead parents to use childcare services that they would otherwise avoid. Within the context of the model, the perceived relative productivity of the mother’s time to childcare of a given quality plays a central role in determining how much of that care parents choose to use. So an optimistic view of childcare quality can lead to adverse developmental outcomes that would not occur with full information. On the other hand, if families underestimate the influence of parental time on children’s human capital, that is precisely when parenting and preschool interventions can have the strongest effects. Thus, a policy allowing mothers to work and develop their careers while at the same time delivering child-oriented

interventions can form the basis of a win-win policy, empowering mothers and improving child development. Parenting and childcare interventions acquire a particular importance in such a world of distorted beliefs.

3 The Effect of Childcare on the Family: A Literature review

We now review the empirical literature on the effects of childcare in low- and middle-income countries. We highlight throughout the few cases where studies evaluate both the effects on mothers and their children, as most focus only on one. Effects on family members other than mothers and their children are even less studied. The model has rationalized various patterns of effects of making childcare less costly and more broadly available. It is possible for mothers to take up childcare but not start working or even reduce their hours of work. In this case, we expect positive effects on child development. It is also possible to observe simultaneous entry into the labor market and take-up of childcare. Here, child development will only improve if the quality of the childcare provided is high enough. Although the model provides us with a way to organize our thoughts, linking the results of the studies to its implications is not straightforward. Despite the model's insights, critical elements, such as quality of childcare or even child outcomes, are often not measured.

Childcare use is often low at baseline, with the exception of certain urban settings (Clark, Kabiru, Laszlo *et al.*, 2019; Beam, Fitzpatrick, and Reimão, 2025; Fang and Miao, 2024). There is much greater variation in baseline maternal work. Where mothers are working, many

often rely on informal care from other household members like grandmothers ([Attanasio, Paes de Barros, Carneiro *et al.*, 2024](#); [Fang and Miao, 2024](#)).

In some low- and middle-income contexts, women are already working and using informal care when childcare interventions are used. [Bassi, Besbas, Dinarte-Diaz *et al.* \(Forthcoming\)](#) evaluate randomized preschool construction paired with parenting education in rural Mozambique, where 98.6% of control mothers worked. Treated children are 6pp (10%) more likely to be enrolled in primary school and 6pp (9%) more likely to be in the correct grade for their age at endline (ages 6-8) with higher literacy (0.169SD), math (0.115SD) and fine motor skills for writing (0.135SD).¹³ Parental child stimulation increased by 0.17SD and mothers also were 1.3pp less likely to work (although this is not significant), perhaps because childcare made maternal time more efficient at increasing children’s human capital.

In Rio de Janeiro, Brazil ([Attanasio, Paes de Barros, Carneiro *et al.*, 2024](#)), where 70% of mothers work at the beginning of the study, randomized access to daycare seats has no effect on maternal participation in the labor force, although the height-for-age and weight-for-age of the children increased. Single mothers in Nairobi, Kenya, many already working and using childcare, reduced their working hours when randomly offered a childcare subsidy, but without a subsequent drop in income ([Clark, Kabiru, Laszlo *et al.*, 2019](#)), suggesting a switch to higher-paid work. Becoming eligible for kindergarten in urban China, where formal childcare use is already common in children’s early years, has no effect on childcare use or work ([Fang and Miao, 2024](#)).

When baseline maternal employment is low, childcare can induce mothers to work ([Clark, Kabiru, Laszlo *et al.*, 2019](#); [Hojman and Lopez Boo, 2022](#); [Berlinski, Galiani, and McEwan,](#)

¹³Treatment had no significant effect on executive function or socio-emotional development however.

2011; Calderón, 2014; Martinez, Naudeau, and Pereira, 2017), although in Uganda the entry effect is confined to single mothers, with no aggregate or married-mother effect (Bjorvatn, Ferris, Gulesci *et al.*, 2025), and in Burkina Faso the probability of salaried work is essentially unchanged while earnings and time in paid work rise (Ajayi, Dao, and Koussoube, 2022). It can also improve children’s development (Martinez, Naudeau, and Pereira, 2017; Hojman and Lopez Boo, 2022; Bjorvatn, Ferris, Gulesci *et al.*, 2025; Ajayi, Dao, and Koussoube, 2022; Jakiela, Ozier, Fernald *et al.*, 2024; Dean and Jayachandran, 2020; Bietenbeck, Ericsson, and Wamalwa, 2019). In Clark, Kabiru, Laszlo *et al.* (2019) the employment response is driven by married women, whose baseline employment is indeed low (50%), in contrast to single women, whose initial participation rate is 79%. In the Gaza district of Mozambique, where only 24% of primary caregivers in control communities worked in a given month, randomized development of community preschools increased the probability of work by 3.7pp and increased an aggregate child-development index (combining ASQ¹⁴ domains and receptive vocabulary) for children by 0.17SD (Martinez, Naudeau, and Pereira, 2017). However, the model identifies several cases in which childcare fails to push mothers to work or to improve children’s development.

Some interventions provide too little coverage. Halim, Johnson, and Perova (2022), using variation in preschool seats across birth cohort and districts in Indonesia, find part-day preschool increases only unpaid work on the family business. They posit that the short program, at only three hours of care, may be to blame. A randomized offer to three hours of after school care in Chile has similarly small effects on maternal employment (Martínez and Perticará, 2017).

¹⁴Ages and Stages Questionnaire.

Even with full-time care, wages may be too low to make work worthwhile. In Uganda, married mothers randomly offered childcare subsidies are no more likely to work. But fathers increase time spent on wage work, perhaps because fathers have higher wages than mothers (Bjorvatn, Ferris, Gulesci *et al.*, 2025). Married mothers are more likely to work when offered an equivalent cash transfer instead, mainly by entering self-employment. Credit constraints may be a limiting factor, though single mothers increase work after receiving a childcare subsidy alone. Increasing household resources alone is insufficient to improve children’s development: childcare subsidies improved children’s school readiness more than equivalent cash grants.

Childcare may be more effective when paired with employment programs. While a paired treatment hasn’t been directly studied relative to childcare alone, Ajayi, Dao, and Koussoube (2022) consider the effect of randomly constructed community care centers that offered slots to participants in a public works program in Burkina Faso. Childcare increased an index of women’s employment outcomes by 0.08SD even a year after the conclusion of the official public works program, driven by salaried income rather than the extensive margin (the probability of salaried work was essentially unchanged), and improved children’s motor skills, but not language skills. The COVID-19 pandemic, however, disrupted the rollout and may have limited the program’s effects.

Norms around maternal work may lower returns to work further, keeping mothers out of the labor force even when time constraints are relieved. See for example, Jalota and Ho (2024) on the strength of social norms in determining employment outcomes for women. In Indonesia and Rajasthan, India, where market work by mothers is uncommon at baseline, childcare does not draw mothers into paid market work: in Indonesia, employment rises

significantly (4.8pp per additional preschool per 1,000 children), but the increase is entirely in unpaid work on the family farm or business (Halim, Johnson, and Perova, 2022); in Rajasthan, randomized daycare centers left overall labor-force participation essentially unchanged, with only a modest 2.3pp increase in the probability of being paid in cash (Nandi, Agarwal, Chandrashekar *et al.*, 2020).

In theory, higher-quality care requires a smaller subsidy to induce work at fixed price $p(q)$ because it is a better substitute for mother’s time: higher-quality care should also improve children’s development more. Hojman and Lopez Boo (2022) find that higher quality care increases parental investment and socio-emotional skills more than low-quality care, and mothers are less likely to work. High-quality providers may affect beliefs on the returns to parents’ time or otherwise make parental time more productive. Consistent with the model we discussed, higher-quality can raise the demand for child human capital more, and because the mother’s time input rises with that demand, a larger part of the household’s response can take the form of parental investment rather than market hours.

Providing quality care can be difficult though. The market for private daycare in informal settlements in Kenya, for example, operates at a low quality, low price equilibrium (Beam, Fitzpatrick, and Reimão, 2025). A quality improvement initiative raises market-wide quality, including among nearby untreated firms, but fails to change prices. Providers may not be able to recoup the cost of quality improvements without intervention. Other interventions designed to improve provider quality have significantly increased children’s language, math, cognition, and executive function skills (Ganimian, Muralidharan, and Walters, 2024; Andrew, Attanasio, Bernal *et al.*, 2024), while Araujo, Carneiro, Cruz-Aguayo *et al.* (2016) find that higher-quality teachers improve children’s math, language, and executive function

skills. Reorganizing provision is not the same as raising its quality: the Colombian transition from home-based care to large centers *lowered* cognitive skills (Bernal, Attanasio, Peña *et al.*, 2019). Indeed, quality matters relative to the counterfactual. Dean and Jayachandran (2020) find private-kindergarten subsidies in India raise cognitive scores most for children unlikely to attend private care otherwise and whose counterfactual was the lower-quality public option, with smaller gains for children likely to attend private care anyway. Where children would otherwise attend school, childcare can have negative (Bouguen, Filmer, Macours *et al.*, 2018) or neutral effects (Berkes and Bouguen, 2022). More well-off families may also have better home environments or better alternative care arrangements, limiting gains from childcare for these children (Blimpo, Carneiro, Jervis *et al.*, 2022; Brinkman, Hasan, Jung *et al.*, 2017).

When care is offered by household members other than the mother, subsidized childcare frees their time as well. In Brazil, Attanasio, Paes de Barros, Carneiro *et al.* (2024) show that randomized access to daycare seats increases grandparents' and older siblings' labor supply. Older siblings aged 10 to 14 in Mozambique living in communities with newly constructed preschools spend 71% more time on schoolwork and 32% less time caring for younger siblings (Martinez, Naudeau, and Pereira, 2017). Providing care for one child also lowers the effective cost of care for all children. Martínez and Perticarà (2017) find randomized access to after-school care increases preschool enrollment for siblings, for example. Initially, sending children to preschool can also change attendance norms or habits. Bassi, Besbas, Dinarte-Diaz *et al.* (Forthcoming) find younger siblings of treated children are more likely to attend preschool in the following years.

The effect on fathers' work is mixed. In China, children reaching the age of eligibility

for kindergarten had no effect on fathers’ work (Fang and Miao, 2024). On the other hand, randomized access to childcare subsidies in Uganda increases fathers’ wage work (Bjorvatn, Ferris, Gulesci *et al.*, 2025) and community care in the Democratic Republic of the Congo increases the likelihood that fathers work in commercial agriculture or as entrepreneurs (Donald, Lowes, and Vaillant, 2024). Using variation in subsidized childcare seats across birth cohorts and location in Mexico, Calderón (2014) finds that fathers were 1.2pp less likely to work for every additional subsidized childcare seat per 10 children in the municipality. In all of these settings, women spend a greater part of their day doing housework and childcare at baseline, though in Uganda parents are closer to parity (about 7 hours per day of unpaid care work for women vs. 5 for men in national time-use data cited by Bjorvatn, Ferris, Gulesci *et al.*, 2025). Only the Mexican study measures job quality directly, and there employed fathers move to better jobs without increasing hours. We observe a shift toward salaried employment in Uganda and toward commercial agriculture and non-agricultural self-employment in the DRC.¹⁵ A natural hypothesis consistent with all three patterns is that childcare frees fathers from providing care on erratic schedules, allowing them to take up less flexible but better-paid work; for Uganda and the DRC this remains a hypothesis rather than an observed response.

Many of the results in the literature are consistent with the model, which can rationalize both the labor supply responses and the different child development impacts, based on the prevailing context and the quality of care offered. Interestingly, in the few studies that measure parental inputs, offers of care and quality improvements are accompanied by increases in parental stimulation and investment, and in some subgroups by more time playing with

¹⁵There is no direct evidence on schedule flexibility, hourly pay, or which activities were displaced.

children, even where maternal work changes little or falls (Bassi, Besbas, Dinarte-Diaz *et al.*, Forthcoming; Blimpo, Carneiro, Jervis *et al.*, 2022; Martinez, Naudeau, and Pereira, 2017; Hojman and Lopez Boo, 2022). These estimates come from bundled interventions and from investment indices rather than from experimental variation in quality alone, so they do not isolate a causal effect of care quality on maternal time; they are, however, what the model’s complementarity between high-quality childcare and maternal time in producing child human capital would predict. The model abstracted from the role of other family members who may serve as informal caregivers. As expected, the literature suggests they too are freed to pursue work or schooling when free childcare is offered.

4 Obstacles to Childcare’s Success in Low and Middle Income Countries

Next, we discuss in greater detail the limitations of childcare as a solution for women’s labor supply, empowerment and children’s development. We focus on four particular cases: social norms, labor market opportunities, counterfactual care, and limited resources to take advantage of early inputs. Many may work in tandem. For example, social norms against maternal work may limit available labor market opportunities for mothers as well as make formal care less common. More research is still needed on all of these as potential limiting factors, in a wider array of settings.

4.1 Norms for Maternal Work

Social norms around whether mothers should work and what work women should do can prevent them from working outside the home. In two extraordinary papers, [Jalota and Ho \(2024\)](#) and [Ho, Jalota, and Karandikar \(2024\)](#) show two key results for India based on field experiments. First, while women are often willing to take up work within the home, they are reluctant to do so outside the home, even if it significantly increases household income and even if it is designed to be in a safe environment close to home, where children can also attend. Second, and importantly, work-from-home jobs act as a gateway to working outside the home. This is an important insight because it reveals there is a dynamic in shifting the norms towards work for women, even in societies where the norms are highly restrictive. An interesting and related question is whether there are strong peer effects from maternal work behavior. For example, suppose that a childcare intervention draws some women living in families with less strict norms. Does this affect other women and does it change the work attitudes of their daughters as suggested in [Kaytan \(2026\)](#)? In any case, the two papers cited above illustrate the difficulty of breaking with norms, even when the availability of jobs is not an issue.

A broad literature reaches similar conclusions. [Boudet, Petesch, Turk *et al.* \(2013\)](#) conduct a global qualitative analysis and find that while paid work may be socially acceptable in many countries, communities often discourage work outside the home or work that interrupts household chores, especially in rural areas. In Indonesia, the setting of [Halim, Johnson, and Perova \(2022\)](#), women in many communities report limited mobility outside the home, especially outside cities. [Halim, Johnson, and Perova \(2022\)](#) find childcare raises the probability

of working, but almost entirely through unpaid work at home. [Boudet, Petesch, Turk *et al.* \(2013\)](#) also suggest norms can be mutable. For example, women may need to contribute by working if their husbands are unemployed. Education may affect norms over a long horizon. For example, increasing how long children stay in school may indirectly improve social norms on maternal work in the long-run. Similarly, by making it easier to work, girls may invest more in their human capital, as shown with after-school care in the Netherlands ([Kaytan, 2026](#)).

Few have studied the effect of childcare on norms directly. Childcare appears to have limited effects on decision making or autonomy in the home ([Ajayi, Dao, and Koussoube, 2022](#); [Clark, Kabiru, Laszlo *et al.*, 2019](#); [Richardson, Harper, Schmitz *et al.*, 2018](#)), although [Chiappori, Meghir, and Okuyama \(2025\)](#) show that the availability of childcare does improve women’s bargaining power, albeit in Japan, making it much less relevant for LMICs. Effects on the incidence of intimate partner violence range from null to modest reductions: randomized childcare subsidies in Uganda left intimate partner violence (IPV) unchanged, in contrast to the equivalent cash grants, which may have increased physical IPV ([Bjorvatn, Ferris, Gulesci *et al.*, 2025](#)), while the Rajasthan daycare trial found some evidence that IPV fell ([Richardson, Harper, Schmitz *et al.*, 2018](#)). That being said, [Fang and Miao \(2024\)](#) find that the effect of children being just eligible for kindergarten on mothers switching to non-agricultural work is greatest in localities with more traditional gender norms. In these localities, informal care may be less commonly used, increasing the impact of formal care.

4.2 Labor Market Opportunities

Both norms and local economic conditions can limit the demand for women’s labor ([Boudet, Petesch, Turk *et al.*, 2013](#)). The kind of work available to women may also not give them control over their wages, which may be necessary to translate labor market participation into power over household choices ([Jayachandran and Voena, 2026](#)).

[Bjorvatn, Ferris, Gulesci *et al.* \(2025\)](#) provide direct evidence of the role of credit versus time constraints for mothers’ work. In their Ugandan sample, mothers are much more likely to be self-employed than in wage labor (33 versus 12 percent at baseline), with the reverse pattern for fathers. Overall, cash grants increased mothers’ self-employment hours substantially (49%), while childcare subsidies did not on average. Looking at subgroups, only the cash grant raised married mothers’ labor supply and business income. For single mothers however, the childcare subsidy raised self-employment and roughly doubled business profits, as expected if credit constraints bind for couples while time constraints are a greater problem for single mothers. Childcare subsidies raised married fathers’ labor income. [Bjorvatn, Ferris, Gulesci *et al.* \(2025\)](#) suggest fathers increase work more than mothers because fathers’ potential wages are much higher than mothers’.

The effect of childcare on formality and type of work is mixed. Formal jobs may not be readily available in some communities, especially for women. In other cases, they may be available, but take up may be low. In Vietnam, mothers whose child was born just after a birth cut-off for preschool (and who therefore are entitled to preschool one year later than children born just before the cut-off) are significantly more likely to work a wage job and less likely to remain in self-employed farm work ([Dang, Hiraga, and Viet Nguyen, 2022](#)). In that

study, the effect of childcare is greater in localities with higher average incomes and for rural villages closer to other towns, suggesting that local economic conditions may affect access to wage work. [Ajayi, Dao, and Koussoube \(2022\)](#) find that randomized community care centers left the probability of doing any salaried work essentially unchanged, but increased salaried earnings and time in paid work: total monthly salaried income increased by 46% and paid work time by 18%, raising a summary index of women’s employment outcomes by 0.08SD ($p < 0.05$). The sample was drawn from participants in a public works program, suggesting the possible advantages of pairing childcare with employment initiatives. On the other hand, single mothers in Nairobi shift from flexible informal work to jobs with regular hours and higher hourly earnings when they access randomized childcare subsidies; married mothers show no such shift, responding instead by entering employment ([Clark, Kabiru, Laszlo *et al.*, 2019](#)). [Halim, Johnson, and Perova \(2022\)](#) find that women are more likely to work where more preschools are available, but only in unpaid work on the family farm or business.

4.3 The Role of Alternative Care Arrangements

Implicitly, the effect of any program is relative to an alternative care option. Alternative care may be provided by mothers, but also by older siblings, other relatives or non-relatives. Older siblings, particularly older sisters, may attend school more infrequently to care for younger siblings ([Boudet, Petesch, Turk *et al.*, 2013](#); [Martinez, Naudeau, and Pereira, 2017](#); [Bassi, Besbas, Dinarte-Diaz *et al.*, Forthcoming](#)). Formal care may be preferable if this is the case, even without a contemporaneous change in maternal labor supply.

In the case where the alternative care option to formal childcare is maternal care, there

can remain variation in the form it takes, often because of variation in parenting approaches and norms. Maternal care may be intensive, providing high levels of child stimulation and nurturing, but it can also be casual, performed simultaneously with work, especially in developing countries where informal work is more common (Delecourt and Fitzpatrick, 2021). Mothers who, absent free childcare, combine work with minding their children are already economically active; for them the opportunity is not labor-market entry but a change in how they work. Consistent with this, Donald, Lowes, and Vaillant (2024) find suggestive evidence that community-based childcare centers reduce mothers’ multi-tasking while farming, increasing plot productivity. Children 6-48 months old with access to community-based childcare centers also had higher CREDI scores.¹⁶ Indeed, as discussed earlier and not surprisingly, childcare is most effective when it offers a richer environment than the child’s alternative care. The gains are greatest when the alternative is a home environment where parents lack the means and knowledge to provide stimulation (Araujo, Ogava, and Souza, 2025; Brinkman, Hasan, Jung *et al.*, 2017); effects are small or even negative when children would otherwise attend school or another form of structured care (Bouguen, Filmer, Macours *et al.*, 2018; Dean and Jayachandran, 2020; Berkes and Bouguen, 2022; Blimpo, Carneiro, Jervis *et al.*, 2022). Bernal, Attanasio, Peña *et al.* (2019) show that children in Colombia who were offered to move from community home daycare to lower-quality centers scored 0.11SD lower on one cognitive assessment ($p < 0.05$), with no significant change on a second cognitive measure or in socio-emotional skills, and a small improvement in nutrition, the one domain in which the centers were better resourced. The pattern is consistent with the view that what matters is the quality of care relative to the alternative, domain by domain, not

¹⁶Caregiver Reported Early Development Instrument.

the attendance of the center per se.

4.4 Taking Advantage of Early Inputs

Even when childcare leads to short-term gains in children’s cognitive and socio-emotional development, long-run returns do not always materialize. In Kenya, [Jakiela, Ozier, Fernald *et al.* \(2024\)](#) find living near a pre-primary school increases receptive vocabulary in the native language and expressive vocabulary at age three, but no effect by age four. Effects of preschool do not always fade, however. Controlling for household fixed effects, [Bietenbeck, Ericsson, and Wamalwa \(2019\)](#) find that children in Kenya, as well as Tanzania, who attend preschool start school later, yet score 0.10SD higher on cognitive tests by ages 13-16. Additionally in Brazil, [Araujo, Ogawa, and Souza \(2025\)](#) find that attending preschool raises fifth-grade scores by 0.26SD (math) and 0.29SD (language); the average effect of the mandatory roll-out across all fifth graders is the more modest 0.033–0.037SD, reflecting incomplete take-up. [Dean and Jayachandran \(2020\)](#) find kindergarten scholarships in Karnataka, India, increase cognition through the end of the first grade, though by roughly half of the initial effect. Preschool also increases on-time and consistent primary school attendance in some settings ([Berlinski, Galiani, and Manacorda, 2008](#); [Bassi, Besbas, Dinarte-Diaz *et al.*, Forthcoming](#); [Martinez, Naudeau, and Pereira, 2017](#)).

Preschool may be most effective when coupled with long-term investments and implemented at scale. In the US context, where we have some evidence, [Johnson and Jackson \(2019\)](#) show that Head Start is more effective when K-12 public school spending is exogenously higher, in line with an explanation for racial differences in Head Start’s effect

proposed by [Currie and Thomas \(2000\)](#). Under-resourced primary schools may erode early investments. However, apparent fade-out may actually reflect indirect treatment of control children. For example, teachers may “teach to the bottom,” leading to control children catching up with treated children. Of course, the fact that control children can catch up with their peers that have been treated earlier is important in itself and points to the possibility of later remediation policies, which are often considered low-return and expensive ([Cunha and Heckman, 2007](#)). Evidence of catch-up is also provided in [Meghir, Attanasio, Jervis *et al.* \(2023\)](#).

Control children may also benefit from exposure to treated peers. Controlling for own earlier test score and school fixed effects, [Neidell and Waldfogel \(2010\)](#) find that a child’s math and reading skills rise with the share of classmates who attended preschool: going from no such classmates to the current enrollment rate of about 46% is worth 0.08SD. [List and Uchida \(2024\)](#) find that children who received preschool have stronger medium-term effects when more of their peers also received preschool. Medium-run effects are greatest when the peers in question shared a preschool classroom, suggesting social networks as an important channel. Preschool programs may then have larger effects on children’s long-term development when implemented at scale.

In Appendix Table 1 we present a summary of some of the most important studies on the effects of childcare on both female labor supply and child outcomes, with developing countries in Panel A and developed countries in Panel B. Finally, in Appendix Table 2 we present a sample of childcare policies across different country groups.

5 Discussion and Conclusion

Childcare can be a useful tool to empower mothers and improve children’s long-term outcomes. When it works well, it can provide multiple benefits: mothers can develop their careers, child development can be improved, and other caregivers such as older siblings can return to school, and grandparents can return to work or better enjoy their retirement.

However, it is no silver bullet for all these problems. Achieving these objectives will probably require a package of policy interventions. Childcare can potentially free mothers to join the labor force, but they may be limited by norms, limited demand for women in the labor market, and low or discriminatory wages. The combination of childcare with initiatives that raise the returns to women’s work may be most effective. Suggestive evidence comes from childcare offered within a public-works employment program, where effects lasted even a year after the conclusion of the employment program ([Ajayi, Dao, and Koussoube, 2022](#)). The one factorial test to date paired a childcare subsidy with an unconditional cash grant and found no complementarity, namely that receiving both had similar effects to the cash grant alone ([Bjorvatn, Ferris, Gulesci *et al.*, 2025](#)), suggesting combined interventions that increase the return to working and not toward extra cash on its own.

Children whose mothers are newly working may be sent to low-quality programs, which ultimately may harm their development. Thus, introducing childcare infrastructure has to be accompanied by attention to quality, offering early education with engaging activities and suitable curriculum and materials. Finally, the evidence on siblings released from childcare duties is mixed: in some settings, they are drawn into work while in others their school enrollment or study time rises.

Achieving high-quality preschool is complex in itself and simple interventions are unlikely to work. Quality includes both structural elements (e.g. curricula, staff characteristics, physical environment, teaching and learning materials) and process (e.g. teacher-child interactions, learning opportunities). Typically, quality childcare requires not only the provision of materials, a developmentally appropriate curriculum, and teacher guides, but also training

of childcare staff and ongoing mentoring and coaching (Egert, Fukkink, and Eckhardt, 2018; Eadie, Page, Levickis *et al.*, 2024; Baker-Henningham, Bowers, Francis *et al.*, 2021).

Teacher training alone, for example, has not in many cases succeeded (Blimpo, Carneiro, Jervis *et al.*, 2022; Özler, Fernald, Kariger *et al.*, 2018; Yoshikawa, Leyva, Snow *et al.*, 2015), but has sometimes been more successful when combined with parenting initiatives (Özler, Fernald, Kariger *et al.*, 2018). However, in other cases, paired treatment may be less effective than teacher training alone or worse than the control (Wolf, 2019; Wolf, Aber, Behrman *et al.*, 2019), suggesting that the specifics of implementation are important. Conversely, increased staffing can improve child outcomes on its own (Ganimian, Muralidharan, and Walters, 2024), but may fail unless it is paired with teacher training: Andrew, Attanasio, Bernal *et al.* (2024) compare control, additional staffing alone, and additional staffing combined with teacher training in Colombian public preschools; there is no training-only arm, so the design identifies the incremental effect of adding training where staffing is present, not a formal staffing–training interaction. Staffing alone leads to teachers spending more time on administrative tasks, while adding teacher training improves teaching quality and students’ cognitive development, a pattern consistent with, though not conclusive, the two inputs being complements. Beam, Fitzpatrick, and Reimão (2025) also demonstrate the difficulty of improving private care in low and middle-income countries. Even when training and certification improve quality at participating providers, prices do not respond similarly, consistent with parents’ low willingness to pay for quality they cannot easily observe; the study’s child-development endline is still to come. Private returns to quality alone may be insufficient to induce providers to improve their quality.

The overarching conclusion one can draw from the literature is that childcare, when provided at high enough quality, can help women build careers and improve child development, but there are important obstacles related to norms and, more broadly, labor market opportunities for women. Childcare on its own is unlikely to be transformative for women in many contexts, but can be a key component in a policy that seeks to improve their outcomes and

those of their children.

References

- Ajayi, K. F., Dao, A., and Koussoubé, M. E. J. (2022). ‘The Effects of Childcare on Women and Children : Evidence from a Randomized Evaluation in Burkina Faso.’ *Policy Research Working Paper Series*.
- Almås, I., Drange, N., Meghir, C., and Zachrisson, H. D. (2026). ‘Early Childcare Attendance and Cognitive Skills in Adolescence.’ NBER Working Paper 35109, National Bureau of Economic Research, Cambridge, MA.
- Andrew, A., Attanasio, O., Augsburg, B., Behrman, J., Day, M., Jervis, P., Meghir, C., and Phimister, A. (2020). ‘Mothers’ Social Networks and Socioeconomic Gradients of Isolation.’ Working Paper 28049, National Bureau of Economic Research.
- Andrew, A., Attanasio, O. P., Bernal, R., Sosa, L. C., Krutikova, S., and Rubio-Codina, M. (2024). ‘Preschool Quality and Child Development.’ *Journal of Political Economy*, vol. 132(7):page 2304–2345. ISSN 0022-3808, 1537-534X.
- Araujo, A., Ogawa, B. T., and Souza, A. P. (2025). ‘The Effects of Universal Early-Childhood Education in Brazil.’ *Journal of Human Capital*, vol. 19(3):page 614–663. ISSN 1932-8575, 1932-8664.
- Araujo, M. C., Carneiro, P., Cruz-Aguayo, Y., and Schady, N. (2016). ‘Teacher Quality and Learning Outcomes in Kindergarten.’ *The Quarterly Journal of Economics*, vol. 131(3):pages 1415–1454. ISSN 00335533, 15314650.
- Attanasio, O., Paes de Barros, R., Carneiro, P., Evans, D. K., Lima, L., Olinto, P., and Schady, N. (2024). ‘Public Childcare, Labor Market Outcomes of Caregivers, and Child Development: Experimental Evidence from Brazil.’ Working Paper 30653, National Bureau of Economic Research, Cambridge, MA.
- Attanasio, O., Cunha, F., and Jervis, P. (2019). ‘Subjective Parental Beliefs: Their Measurement and Role.’ Working Paper 26516, National Bureau of Economic Research, Cambridge, MA.
- Baker, M., Gruber, J., and Milligan, K. (2008). ‘Universal Child Care, Maternal Labor Supply, and Family Well-Being.’ *Journal of Political Economy*, vol. 116(4):pages 709–745. ISSN 0022-3808, 1537-534X.
- Baker-Henningham, H., Bowers, M., Francis, T., Vera-Hernández, M., and Walker, S. P. (2021). ‘The Irie Classroom Toolbox, a Universal Violence-Prevention Teacher-Training Programme, in Jamaican Preschools: A Single-Blind, Cluster-Randomised Controlled Trial.’ *The Lancet Global Health*, vol. 9(4):pages e456–e468.

- Bassi, M., Besbas, B., Dinarte-Diaz, L., Ravindran, S., and Reynoso, A. (Forthcoming). ‘From Access to Achievement: The Primary School-Age Impacts of an At-Scale Preschool Construction Program in Highly Deprived Communities.’ *American Economic Journal: Applied Economics*. ISSN 1945-7782.
- Beam, E. A., Fitzpatrick, A., and Reimão, M. E. (2025). ‘Strengthening Childcare Markets through Social Franchising: Evidence from Kenya.’ G2LM|LIC Policy Brief 82, IZA – Institute of Labor Economics, Gender, Growth and Labour Markets in Low-Income Countries (G2LM|LIC) Programme. Preliminary results from an ongoing cluster-randomized trial; endline, including child development outcomes, expected 2026.
- Becker, G. S. (1965). ‘A Theory of the Allocation of Time.’ *Economic Journal*, vol. 75(299):pages 493–517.
- Benjamin, D. (1992). ‘Household Composition, Labor Markets, and Labor Demand: Testing for Separation in Agricultural Household Models.’ *Econometrica*, vol. 60(2):pages 287–322.
- Berkes, J. and Bouguen, A. (2022). ‘Estimating Preschool Impacts when Counterfactual Enrollment Varies: Bounds, Conditional LATE and Machine Learning.’ *Center for Effective Global Action Working Paper*, (88).
- Berlinski, S., Galiani, S., and Manacorda, M. (2008). ‘Giving children a better start: Preschool attendance and school-age profiles.’ *Journal of Public Economics*, vol. 92(5):pages 1416–1440. ISSN 0047-2727.
- Berlinski, S., Galiani, S., and McEwan, P. J. (2011). ‘Preschool and Maternal Labor Market Outcomes: Evidence from a Regression Discontinuity Design.’ *Economic Development and Cultural Change*, vol. 59(2):pages 313–344. ISSN 0013-0079, 1539-2988.
- Bernal, R., Attanasio, O., Peña, X., and Vera-Hernández, M. (2019). ‘The effects of the transition from home-based childcare to childcare centers on children’s health and development in Colombia.’ *Early Childhood Research Quarterly*, vol. 47:pages 418–431. ISSN 0885-2006.
- Bietenbeck, J., Ericsson, S., and Wamalwa, F. M. (2019). ‘Preschool Attendance, Schooling, and Cognitive Skills in East Africa.’ *Economics of Education Review*, vol. 73.
- Bjorvatn, K., Ferris, D., Gulesci, S., Nasgowitz, A., Somville, V., and Vandewalle, L. (2025). ‘Childcare, Labor Supply, and Business Development: Experimental Evidence from Uganda.’ *American Economic Journal: Applied Economics*, vol. 17(2):page 75–101. ISSN 1945-7782.
- Blimpo, M. P., Carneiro, P., Jervis, P., and Pugatch, T. (2022). ‘Improving Access and Quality in Early Childhood Development Programs: Experimental Evidence from the Gambia.’ *Economic Development and Cultural Change*, vol. 70(4):pages 1479–1529.
- Blundell, R., Chiappori, P., and Meghir, C. (2005). ‘Collective Labor Supply with Children.’ *Journal of Political Economy*, vol. 113(6):pages 1277–1306. ISSN 0022-3808, 1537-534X.

- Boudet, A. M. M., Petesch, P., Turk, C., and Thumala, A. (2013). ‘On Norms and Agency : Conversations about Gender Equality with Women and Men in 20 Countries.’ *World Bank Publications - Books*.
- Bouguen, A., Filmer, D., Macours, K., and Naudeau, S. (2018). ‘Preschool and Parental Response in a Second Best World: Evidence from a School Construction Experiment.’ *Journal of Human Resources*, vol. 53(2):pages 474–512. ISSN 0022-166X, 1548-8004.
- Bowen, H. R. (1943). ‘The Interpretation of Voting in the Allocation of Economic Resources.’ *Quarterly Journal of Economics*, vol. 58(1):pages 27–48.
- Brinkman, S. A., Hasan, A., Jung, H., Kinnell, A., and Pradhan, M. (2017). ‘The Impact of Expanding Access to Early Childhood Education Services in Rural Indonesia.’ *Journal of Labor Economics*, vol. 35(S1):pages S305–S335. ISSN 0734-306X, 1537-5307.
- Calderón, G. (2014). ‘The effects of child care provision in Mexico.’ Working Paper 2014–07, Banco de México.
- Chiappori, P.-A. (1988). ‘Rational household labor supply.’ *Econometrica: Journal of the Econometric Society*, pages 63–90.
- Chiappori, P.-A. (1992). ‘Collective labor supply and welfare.’ *Journal of Political Economy*, pages 437–467.
- Chiappori, P.-A., Costa-Dias, M., and Meghir, C. (2018). ‘The Marriage Market, Labor Supply, and Education Choice.’ *Journal of Political Economy*, vol. 126(S1):pages S26–S72. ISSN 0022-3808. Publisher: The University of Chicago Press.
- Chiappori, P.-A., Meghir, C., and Okuyama, Y. (2025). ‘Intra-Household Welfare Inequality and Household Public Goods.’ NBER Working Paper 32645, National Bureau of Economic Research. Revised February 2025. Also Cowles Foundation Discussion Paper 2395R1. Previously circulated as “Intrahousehold Welfare: Theory and Application to Japanese Data”.
- Clark, S., Kabiru, C. W., Laszlo, S., and Muthuri, S. (2019). ‘The Impact of Childcare on Poor Urban Women’s Economic Empowerment in Africa.’ *Demography*, vol. 56(4):pages 1247–1272. ISSN 1533-7790.
- Cunha, F., Elo, I., and Culhane, J. (2022). ‘Maternal Subjective Expectations about the Technology of Skill Formation Predict Investments in Children One Year Later.’ *Journal of Econometrics*, vol. 231(1):pages 3–32.
- Cunha, F. and Heckman, J. (2007). ‘The Technology of Skill Formation.’ vol. 97(2):page 81.
- Currie, J. and Thomas, D. (2000). ‘School Quality and the Longer-Term Effects of Head Start.’ *Journal of Human Resources*, vol. 35(4):pages 755–774.
- Dang, H.-A. H., Hiraga, M., and Viet Nguyen, C. (2022). ‘Childcare and maternal employment: Evidence from Vietnam.’ *World Development*, vol. 159:page 106022. ISSN 0305-750X.

- Dean, J. T. and Jayachandran, S. (2020). ‘Attending kindergarten improves cognitive but not socioemotional development in India.’
- Delecourt, S. and Fitzpatrick, A. (2021). ‘Childcare Matters: Female Business Owners and the Baby-Profit Gap.’ *Management Science*, vol. 67(7):pages 4455–4474.
- Devercelli, A. E. and Beaton-Day, F. (2020). ‘Better Jobs and Brighter Futures: Investing in Childcare to Build Human Capital.’ Tech. rep., World Bank, Washington DC.
- Donald, A., Lowes, S., and Vaillant, J. (2024). ‘Experimental Evidence on Rural Childcare Provision.’
- Eadie, P., Page, J., Levickis, P., Elek, C., Murray, L., Wang, L., and Lloyd-Johnsen, C. (2024). ‘Domains of Quality in Early Childhood Education and Care: A Scoping Review of the Extent and Consistency of the Literature.’ *Educational Review*, vol. 76(4):pages 1057–1086.
- Egert, F., Fukkink, R. G., and Eckhardt, A. G. (2018). ‘Impact of In-Service Professional Development Programs for Early Childhood Teachers on Quality Ratings and Child Outcomes: A Meta-Analysis.’ *Review of Educational Research*, vol. 88(3):pages 401–433.
- Evans, D. K., Jakiela, P., and Knauer, H. A. (2021). ‘The impact of early childhood interventions on mothers.’ *Science*, vol. 372(6544):pages 794–796. ISSN 0036-8075, 1095-9203.
- Fang, G. and Miao, L. (2024). ‘Expanding boundaries: The Impact of kindergarten availability on women’s employment in China.’ *Labour Economics*, vol. 88:page 102542. ISSN 0927-5371.
- Ganimian, A. J., Muralidharan, K., and Walters, C. R. (2024). ‘Augmenting State Capacity for Child Development: Experimental Evidence from India.’ *Journal of Political Economy*, vol. 132(5). Earlier version: NBER Working Paper 28780, 2021, <https://doi.org/10.3386/w28780>.
- Gronau, R. (1977). ‘Leisure, Home Production, and Work—the Theory of the Allocation of Time Revisited.’ *Journal of Political Economy*, vol. 85(6):pages 1099–1123.
- Halim, D., Johnson, H. C., and Perova, E. (2022). ‘Preschool Availability and Women’s Employment: Evidence from Indonesia.’ *Economic Development and Cultural Change*, vol. 71(1):pages 39–61. ISSN 0013-0079, 1539-2988.
- Heath, R. and Mobarak, M. (2015). ‘Manufacturing growth and the lives of Bangladeshi women.’ *Journal of Development Economics*, vol. 115:pages 1–15. ISSN 0304-3878.
- Ho, L., Jalota, S., and Karandikar, A. (2024). ‘Bringing Work Home: Flexible Work Arrangements as Gateway Jobs for Women in West Bengal.’ Working paper, revise and resubmit, *The Quarterly Journal of Economics*.
- Hojman, A. and Lopez Boo, F. (2022). ‘Public childcare benefits children and mothers: Evidence from a nationwide experiment in a developing country.’ *Journal of Public Economics*, vol. 212:page 104686. ISSN 0047-2727.

- Jakiela, P., Ozier, O., Fernald, L. C. H., and Knauer, H. A. (2024). ‘Preprimary Education and Early Childhood Development: Evidence from Government Schools in Rural Kenya.’ *Journal of Development Economics*, vol. 171. Article 103340.
- Jalota, S. and Ho, L. (2024). ‘What Works for Her? How Work-from-Home Digital Jobs Affect Female Labor Force Participation in Urban India.’ Working paper, SSRN 4739387.
- Jayachandran, S. and Voena, A. (2026). ‘Women’s Power in the Household.’ *Journal of Economic Literature*.
- Johnson, R. C. and Jackson, C. K. (2019). ‘Reducing Inequality through Dynamic Complementarity: Evidence from Head Start and Public School Spending.’ *American Economic Journal: Economic Policy*, vol. 11(4):pages 310–349.
- Kaytan, S. (2026). ‘The Long-term Effects of After-school Care.’ *Job Market Paper*.
- Lindahl, E. (1958). ‘Just Taxation—A Positive Solution.’ In R. A. Musgrave and A. T. Peacock, eds., *Classics in the Theory of Public Finance*, pages 168–176. Macmillan, London. Translated excerpt from *Die Gerechtigkeit der Besteuerung* (1919).
- List, J. A. and Uchida, H. (2024). ‘Here Today, Gone Tomorrow? Toward an Understanding of Fade-out in Early Childhood Education Programs.’ Working Paper 33027, National Bureau of Economic Research, Cambridge, MA.
- Lundberg, S. and Pollak, R. A. (1993). ‘Separate Spheres Bargaining and the Marriage Market.’ *Journal of Political Economy*, vol. 101(6):pages 988–1010. ISSN 0022-3808. Publisher: The University of Chicago Press.
- Martinez, S., Naudeau, S., and Pereira, V. A. (2017). ‘Preschool and Child Development Under Extreme Poverty: Evidence from a Randomized Experiment in Rural Mozambique.’ Working Paper 3092440, SSRN, Rochester, NY.
- Martínez, C. A. and Perticará, M. (2017). ‘Childcare effects on maternal employment: Evidence from Chile.’ *Journal of Development Economics*, vol. 126:pages 127–137. ISSN 0304-3878.
- Mazzocco, M. (2007). ‘Household intertemporal behaviour: A collective characterization and a test of commitment.’ *The Review of Economic Studies*, vol. 74(3):pages 857–895.
- Meghir, C., Attanasio, O., Jervis, P., Day, M., Makkar, P., Behrman, J., Gupta, P., Pal, R., Phimister, A., Vernekar, N., and Grantham-McGregor, S. (2023). ‘Early Stimulation and Enhanced Preschool: A Randomized Trial.’ *Pediatrics*, vol. 151(Supplement 2):page e2023060221H. ISSN 0031-4005.
- Molina, T. and Tanaka, M. (2023). ‘Globalization and Female Empowerment: Evidence from Myanmar.’ *Economic Development and Cultural Change*, vol. 71(2):pages 519–565. ISSN 0013-0079, 1539-2988.

- Nandi, A., Agarwal, P., Chandrashekar, A., and Harper, S. (2020). ‘Access to affordable daycare and women’s economic opportunities: evidence from a cluster-randomised intervention in India.’ *Journal of Development Effectiveness*, vol. 12(3):pages 219–239. ISSN 1943-9342, 1943-9407.
- Neidell, M. and Waldfogel, J. (2010). ‘Cognitive and Noncognitive Peer Effects in Early Education.’ *Review of Economics and Statistics*, vol. 92(3):pages 562–576.
- Özler, B., Fernald, L. C. H., Kariger, P., McConnell, C., Neuman, M., and Fraga, E. (2018). ‘Combining Pre-School Teacher Training with Parenting Education: A Cluster-Randomized Controlled Trial.’ *Journal of Development Economics*, vol. 133:pages 448–467.
- Richardson, R. A., Harper, S., Schmitz, N., and Nandi, A. (2018). ‘The effect of affordable daycare on women’s mental health: Evidence from a cluster randomized trial in rural India.’ *Social Science & Medicine*, vol. 217:pages 32–41. ISSN 0277-9536.
- Samuelson, P. A. (1954). ‘The Pure Theory of Public Expenditure.’ *The Review of Economics and Statistics*, vol. 36(4):pages 387–389.
- Singh, I., Squire, L., and Strauss, J., eds. (1986). *Agricultural Household Models: Extensions, Applications, and Policy*. Johns Hopkins University Press for the World Bank, Baltimore, MD.
- Voena, A. (2015). ‘Yours, Mine, and Ours: Do Divorce Laws Affect the Intertemporal Behavior of Married Couples?’ *American Economic Review*, vol. 105(8):pages 2295–2332. ISSN 0002-8282.
- Wolf, S. (2019). ‘Year 3 follow-up of the ‘Quality Preschool for Ghana’ interventions on child development.’ *Developmental Psychology*, vol. 55(12):page 2587–2602. ISSN 1939-0599, 0012-1649.
- Wolf, S., Aber, J. L., Behrman, J. R., and Tsinigo, E. (2019). ‘Experimental Impacts of the “Quality Preschool for Ghana” Interventions on Teacher Professional Well-being, Classroom Quality, and Children’s School Readiness.’ *Journal of Research on Educational Effectiveness*, vol. 12(1):page 10–37. ISSN 1934-5747, 1934-5739.
- World Bank (2025). ‘Labor force participation rate (% of population).’ Data table, World Bank Gender Data Portal.
- Yoshikawa, H., Leyva, D., Snow, C. E., Treviño, E., Barata, M. C., Weiland, C., Gomez, C. J., Moreno, L., Rolla, A., D’Sa, N., and Arbour, M. C. (2015). ‘Experimental Impacts of a Teacher Professional Development Program in Chile on Preschool Classroom Quality and Child Outcomes.’ *Developmental Psychology*, vol. 51(3):pages 309–322.